

## **COURSE ASSIGNMENT**

### **Assignment Information :**

| <b>Particular</b>             | <b>Details</b>                                                                                                                                                                           |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Department:                   | IT                                                                                                                                                                                       |
| Programme:                    | B.TECH                                                                                                                                                                                   |
| Course Code & Course Name     | CSA0703-Computer Networks                                                                                                                                                                |
| Academic Year / Batch         | 2026-2027                                                                                                                                                                                |
| Faculty Name                  | Dr.R.Samuel Rajesh Babu                                                                                                                                                                  |
| Assignment Title              | Design and Simulate a Small Enterprise Network Consisting of Multiple Interconnected LANs                                                                                                |
| Date of Issue                 | 01-09-2026                                                                                                                                                                               |
| Date of Submission            | 02-09-2026                                                                                                                                                                               |
| Maximum Marks                 | 100                                                                                                                                                                                      |
| Course Outcome(s) – CO        | Apply networking concepts to design, configure, and analyze computer networks                                                                                                            |
| Bloom's Taxonomy Level        | L4 – Analyze                                                                                                                                                                             |
| SDG Mapping (SDG 1-17)        | SDG 9 – Industry, Innovation and Infrastructure                                                                                                                                          |
| Industry / Societal Relevance | Enterprise network design and simulation are directly relevant to organizational communication, resource sharing, network infrastructure development, and reliable digital connectivity. |

**ASSINGMENT DONE BY:**

SIVARANGJENII B(192521375)

## **B. Assignment Problem / Challenge**

Design and simulate a small enterprise network consisting of multiple interconnected LANs. Apply the functions of the OSI layers to ensure reliable end-to-end data transmission, configure an appropriate IP addressing and subnetting scheme with basic routing between the LANs using a network simulation tool such as Cisco Packet Tracer, and apply TCP, UDP, HTTP, and DNS protocols to demonstrate network communication. Configure and test the network under different traffic conditions, and compare the observed performance of TCP and UDP in terms of packet delivery, delay, and throughput. Explain how the OSI layers, IP addressing/subnetting, routing, and transport/application-layer protocols work together to achieve efficient and reliable communication.

### **1. Problem Statement and Problem Formulation**

A small enterprise network consists of multiple departments or LANs that need to communicate with each other. Each department can have its own LAN containing computers, printers, servers, and other network devices.

For example, an enterprise may have:

- Administration LAN
- Sales LAN
- Technical LAN

Each LAN is connected using a switch, and routers are used to connect the different LANs. Since every LAN has a different network address, routing is required for communication between them.

Cisco Packet Tracer can be used to design, configure, simulate, and analyze this network without requiring physical networking equipment.

The experiment also demonstrates how the different OSI layers work together with IP addressing, subnetting, routing, transport protocols, and application protocols to provide end-to-end communication.

### **2. Objective and Expected Outcomes**

#### **Objective**

The main objective of this assignment is to design and simulate a small enterprise network consisting of multiple interconnected LANs using Cisco Packet Tracer. The assignment aims to apply networking concepts such as IP addressing, subnetting, routing, and the functions of the OSI layers to establish reliable end-to-end communication between different networks.

The assignment also aims to configure and demonstrate TCP, UDP, HTTP, and DNS protocols and to understand their roles in network communication. The network is tested under different traffic conditions to compare TCP and UDP performance based on packet delivery, delay, packet loss, and throughput.

## Expected Outcomes

After completing the assignment, the following outcomes are expected:

- Successfully design and simulate multiple interconnected LANs using Cisco Packet Tracer.
- Configure suitable IP addressing and subnetting schemes for the different LANs.
- Configure basic routing to enable communication between interconnected networks.
- Understand and apply the functions of the OSI layers during end-to-end data transmission.
- Demonstrate communication using TCP and UDP protocols.
- Configure and verify HTTP and DNS services in the simulated network.
- Test network communication under different traffic conditions.
- Compare TCP and UDP based on packet delivery, delay, packet loss, and throughput.
- Analyze the observed network performance and understand the trade-offs between reliable and efficient communication.
- Demonstrate how OSI layers, IP addressing, subnetting, routing, and transport/application-layer protocols work together to achieve effective network communication.

## 3. Requirements, Constraints and Assumptions

### Network Requirements

The following devices can be used:

- 3 Routers
- 3 Switches
- 6 or more PCs
- 1 DNS Server
- 1 HTTP/Web Server
- Ethernet cables
- Serial or Ethernet connections between routers, depending on the topology

A simple arrangement can be:

*Administration PCs → Switch 1 → Router 1*

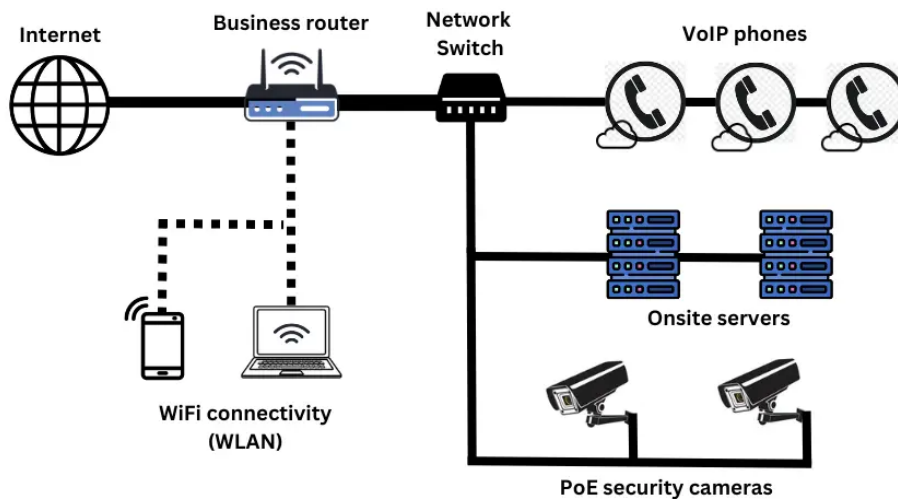
*Sales PCs → Switch 2 → Router 2*

*Technical PCs → Switch 3 → Router 3*

*Router 1 ↔ Router 2 ↔ Router 3*

The routers provide communication between the different LANs.

## A small business LAN



**Figure 1: Small Enterprise Network with Multiple Interconnected LANs**

### Example Network Topology

The network can be designed as follows:

LAN 1: Administration

*PC1, PC2 → Switch 1 → Router 1*

LAN 2: Sales

*PC3, PC4 → Switch 2 → Router 2*

LAN 3: Technical

*PC5, PC6 → Switch 3 → Router 3*

The routers are interconnected so that packets can travel from one LAN to another.

### Purpose of Each Device

#### PC

A PC acts as an end device that generates and receives network data.

#### Switch

A switch connects multiple devices within the same LAN. It uses MAC addresses to forward Ethernet frames to the appropriate device.

## Router

A router connects different networks or subnets. It examines destination IP addresses and forwards packets toward the appropriate network.

## DNS Server

The DNS server converts domain names into IP addresses.

## HTTP Server

The HTTP server provides web pages or web services to client devices.

## IP Addressing

Each device in the network must have a unique IP address.

A private IPv4 address range such as:

192.168.1.0/24

can be divided into smaller subnets using subnetting.

Subnetting divides one large network into multiple smaller networks.

## Subnetting Scheme

For example, the network 192.168.1.0/24 can be divided using a /26 subnet mask.

Subnet mask:

255.255.255.192

This provides four possible subnets.

The addressing scheme can be:

| LAN            | Network Address  | Subnet Mask     | Default Gateway |
|----------------|------------------|-----------------|-----------------|
| Administration | 192.168.1.0/26   | 255.255.255.192 | 192.168.1.1     |
| Sales          | 192.168.1.64/26  | 255.255.255.192 | 192.168.1.65    |
| Technical      | 192.168.1.128/26 | 255.255.255.192 | 192.168.1.129   |

The remaining subnet can be used for future expansion or other network requirements.

## Host Address Range

For Administration LAN:

- Network address: 192.168.1.0
- First usable address: 192.168.1.1
- Last usable address: 192.168.1.62
- Broadcast address: 192.168.1.63

For Sales LAN:

- Network address: 192.168.1.64
- First usable address: 192.168.1.65
- Last usable address: 192.168.1.126
- Broadcast address: 192.168.1.127

For Technical LAN:

- Network address: 192.168.1.128
- First usable address: 192.168.1.129
- Last usable address: 192.168.1.190
- Broadcast address: 192.168.1.191

### **Example PC Configuration**

Administration PC1:

- IP address: 192.168.1.10
- Subnet mask: 255.255.255.192
- Default gateway: 192.168.1.1
- DNS server: 192.168.1.10 or the configured DNS server address

Sales PC:

- IP address: 192.168.1.70
- Subnet mask: 255.255.255.192
- Default gateway: 192.168.1.65

Technical PC:

- IP address: 192.168.1.140
- Subnet mask: 255.255.255.192
- Default gateway: 192.168.1.129

Every device must have a unique IP address within its subnet.

### **Importance of the Default Gateway**

The default gateway is the router interface through which a device communicates with another network.

For example, if a PC in Administration LAN wants to communicate with a PC in Sales LAN, the destination is outside the Administration subnet.

Therefore:

*PC → Default Gateway → Router → Destination Network*

Without a correctly configured default gateway, communication with other networks will fail.

### Router Interface Configuration

Each router interface connected to a LAN is configured with an IP address belonging to that LAN.

For example:

Router 1:

- LAN interface: 192.168.1.1/26

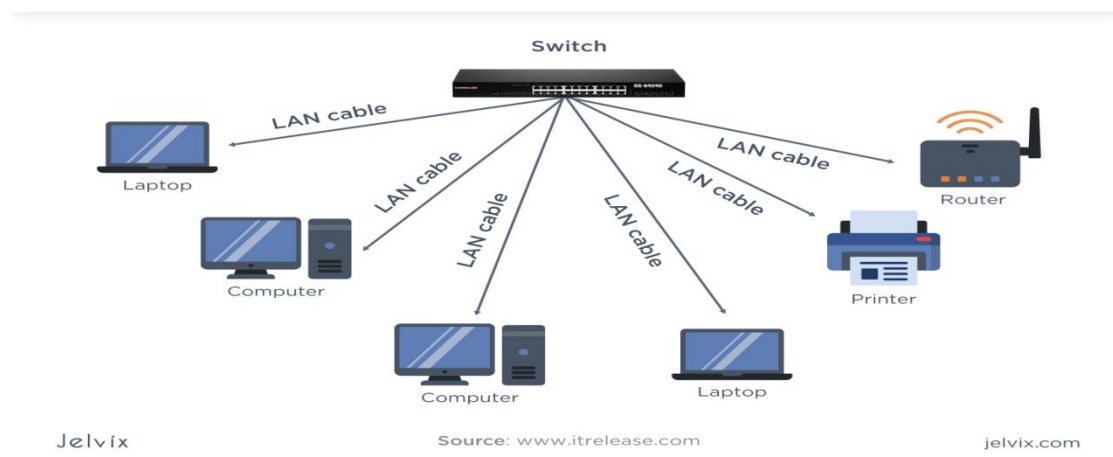
Router 2:

- LAN interface: 192.168.1.65/26

Router 3:

- LAN interface: 192.168.1.129/26

The interfaces connecting the routers are also assigned suitable IP addresses from separate transit networks.



**Figure**

## 2: LAN with Switch, Router, Computers, Laptops, and Printer

### Routing Configuration

Routing is required because different LANs belong to different IP networks.

Routing determines how packets should travel from the source network to the destination network.

Two basic approaches can be used:

- Static routing
- Dynamic routing

For a small network, static routing can be used because it is simple and easy to understand.

Dynamic routing protocols such as RIP or OSPF can also be used for larger or more complex networks.

### **Static Routing Concept**

Suppose a PC in Administration LAN sends data to a PC in Sales LAN.

The source PC sends the packet to Router 1. Router 1 checks its routing table. If the Sales network is reachable through Router 2, Router 1 forwards the packet to Router 2.

Router 2 then forwards the packet toward the destination PC.

The basic path is:

*Source PC → Switch → Router 1 → Router 2 → Switch → Destination PC*

### **Routing Table**

A router maintains a routing table containing information about available networks and the paths used to reach them.

The routing table may contain:

- Destination network
- Subnet mask
- Next-hop address
- Outgoing interface
- Route source

When a packet arrives, the router examines the destination IP address and selects the appropriate route.

## **4. Application of Relevant Course Knowledge / Concepts**

### **Packet Flow Between LANs**

Consider PC1 in Administration LAN communicating with PC3 in Sales LAN.

1. PC1 creates data for PC3.
2. The transport layer adds TCP or UDP information.
3. The network layer adds the source and destination IP addresses.
4. The data-link layer creates an Ethernet frame.
5. The frame is sent to the switch.



6. The switch forwards it toward the router.
7. The router examines the destination IP address.
8. The router consults its routing table.
9. The packet is forwarded toward the Sales LAN.
10. The destination router sends the packet to the appropriate switch.
11. The switch delivers the frame to PC3.
12. The receiving device processes the data through the OSI layers.

## **OSI Layer Functions**

The OSI model contains seven layers. Each layer performs a specific function during communication.

### **Layer 7 – Application Layer**

The Application Layer provides services directly to user applications.

Protocols used in this experiment include:

- HTTP
- DNS

HTTP is used for web communication, while DNS is used for domain-name resolution.

Example:

A user enters:

[www.example.com](http://www.example.com)

The DNS service finds the corresponding IP address.

### **Layer 6 – Presentation Layer**

The Presentation Layer deals with the representation and formatting of data.

Its functions include:

- Data translation
- Encryption and decryption
- Data compression
- Format conversion

It ensures that data is represented in a form that the receiving application can understand.

### **Layer 5 – Session Layer**

The Session Layer establishes, manages, and terminates communication sessions between applications.

It helps maintain communication between applications during a session.

## **Layer 4 – Transport Layer**

The Transport Layer provides end-to-end communication between applications.

The main protocols considered are:

- TCP
- UDP

TCP provides reliable communication, while UDP provides faster communication with less overhead.

## **Layer 3 – Network Layer**

The Network Layer is responsible for logical addressing and routing.

The main protocol is IP.

Functions include:

- Logical addressing
- Packet forwarding
- Routing
- Path selection

Routers operate mainly at this layer.

## **Layer 2 – Data Link Layer**

The Data Link Layer provides communication between devices on the same local network.

It uses:

- MAC addresses
- Frames
- Error detection

Switches primarily operate at this layer.

## **Layer 1 – Physical Layer**

The Physical Layer transmits raw binary bits through the physical medium.

Examples include:

- Ethernet cables
- Network interfaces
- Electrical signals
- Optical signals
- Wireless radio signals

## **TCP Protocol**

TCP stands for Transmission Control Protocol.

TCP is a connection-oriented transport-layer protocol.

Before transmitting data, TCP establishes a connection between the sender and receiver.

TCP provides:

- Reliable delivery
- Ordered delivery
- Error detection
- Acknowledgements
- Flow control
- Retransmission of lost data
- Congestion control

If a packet is lost, TCP can retransmit it.

Therefore, TCP is suitable when data accuracy and reliable delivery are important.

Examples include:

- HTTP/HTTPS
- File transfer
- Email communication
- Web applications

### **TCP Working**

The basic TCP communication process is:

1. Connection establishment
2. Data transmission
3. Acknowledgement
4. Retransmission if necessary
5. Connection termination

TCP uses acknowledgements to confirm successful delivery.

This reliability introduces additional overhead and can increase delay.

### **UDP Protocol**

UDP stands for User Datagram Protocol.

UDP is a connectionless transport-layer protocol.

It does not establish a connection before transmitting data.

UDP does not provide:

- Guaranteed delivery
- Acknowledgements
- Retransmission
- Guaranteed ordering

However, UDP has lower overhead and can provide faster transmission.

UDP is useful for:

- Real-time communication
- Video streaming
- Voice communication
- Online gaming
- DNS queries
- Applications where low delay is important

## **HTTP Protocol**

HTTP stands for HyperText Transfer Protocol.

HTTP operates at the Application Layer.

It is used for communication between web clients and web servers.

For example:

*Client → HTTP Request → Web Server*

*Web Server → HTTP Response → Client*

In Cisco Packet Tracer, an HTTP server can be configured and a PC can access it using the web browser.

Example:

The user enters the server IP address into the browser.

The browser sends an HTTP request, and the server returns the requested web page.

## **DNS Protocol**

DNS stands for Domain Name System.

DNS converts human-readable domain names into IP addresses.

For example:

[www.example.com](http://www.example.com)

may be mapped to:

192.168.1.20

Without DNS, users would need to remember IP addresses instead of domain names.

The basic process is:

*Client → DNS Query → DNS Server*

*DNS Server → IP Address → Client*

The client can then communicate with the required server using the returned IP address.

### **Testing HTTP Communication**

To test HTTP:

1. Configure a server in Packet Tracer.
2. Assign an IP address to the server.
3. Enable the HTTP service.
4. Configure a client PC.
5. Open the web browser on the PC.
6. Enter the web server's IP address or domain name.
7. Observe the HTTP request and response.
8. Verify that the web page is displayed.

Successful web-page access confirms that the required network connectivity and HTTP service are working.

### **Testing DNS Communication**

To test DNS:

1. Configure a DNS server.
2. Assign an IP address to the DNS server.
3. Enable DNS service.
4. Add a domain name and corresponding IP address.
5. Configure the client with the DNS server address.
6. Enter the domain name from the client.
7. Verify that the name is resolved into the correct IP address.

### **Network Testing Using Ping**

The ping command can be used to verify connectivity.

For example:

PC1 can ping:

- Its default gateway
- Another PC in the same LAN

- A PC in another LAN
- The HTTP server
- The DNS server

If the ping is successful, the devices can communicate at the IP level.

If ping fails, possible causes include:

- Incorrect IP address
- Incorrect subnet mask
- Incorrect default gateway
- Incorrect routing
- Interface shutdown
- Incorrect cable connection
- Firewall or service configuration issues

### **Packet Tracer Simulation Mode**

Packet Tracer provides Simulation Mode to observe how packets move through the network.

In Simulation Mode, different protocols can be observed as packets travel through:

*PC → Switch → Router → Router → Switch → Destination PC*

The protocol information and OSI-layer processing can be examined at each stage.

This helps understand the actual movement of data through the network.

### **Traffic Conditions**

The network can be tested under different traffic conditions.

Three basic conditions can be considered:

#### **Low Traffic**

Only a small amount of data is transmitted.

Expected behavior:

- Low delay
- Very low packet loss
- Good throughput
- TCP and UDP both perform well

#### **Moderate Traffic**

Multiple devices generate network traffic simultaneously.

Expected behavior:

- Increased network utilization
- Slight increase in delay
- TCP adjusts its transmission rate
- UDP continues transmitting without waiting for acknowledgements

### Heavy Traffic

A large amount of data is transmitted simultaneously.

Expected behavior:

- Higher congestion
- Increased delay
- Possible packet loss
- TCP reduces its sending rate when congestion is detected
- UDP may experience greater packet loss because it does not provide congestion control in the same way TCP does

## 5. Design / Proposed Solution / Methodology

### Performance Parameters

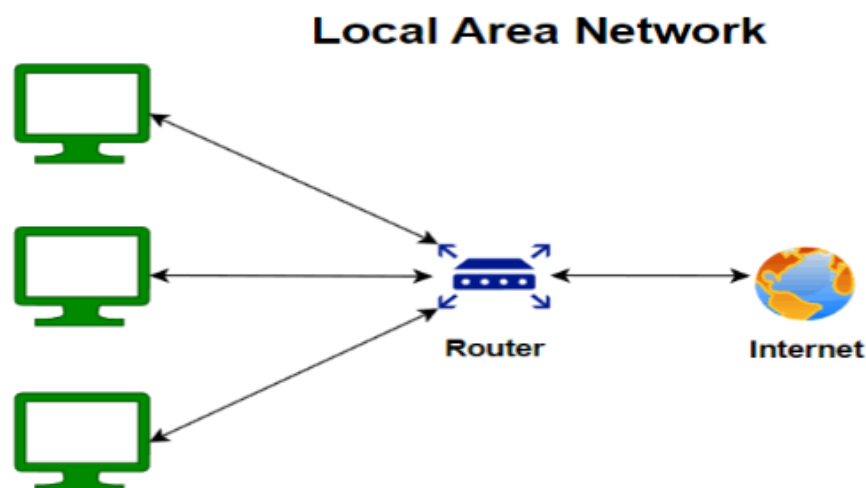
The performance of TCP and UDP can be compared using several parameters.

### Packet Delivery

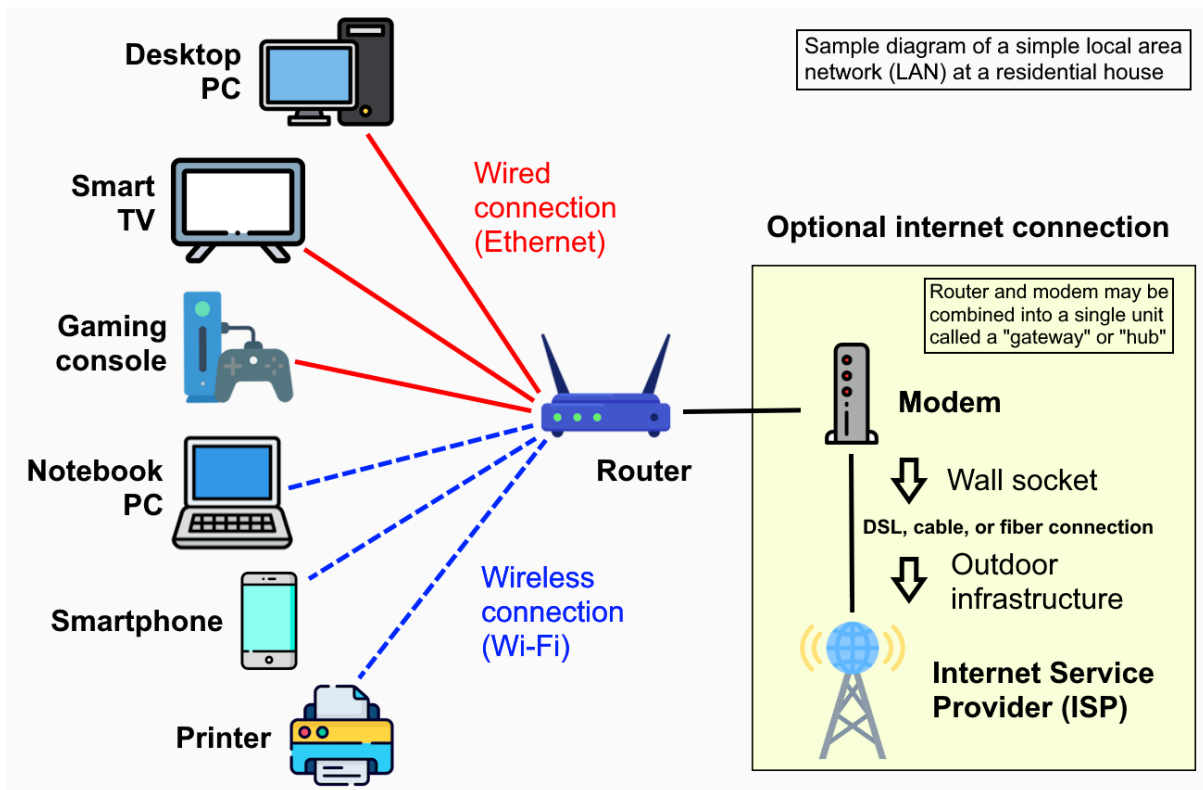
Packet delivery refers to the successful arrival of transmitted packets at the destination.

TCP generally provides reliable delivery because lost packets can be retransmitted.

UDP does not guarantee delivery, so packets may be lost without retransmission.



**Figure 3: Small Business LAN Network Architecture**



**Figure 4: Local Area Network with Router and Internet Connection**

## Delay

Delay is the time required for data to travel from the source to the destination.

TCP may have higher delay because it uses connection establishment, acknowledgements, retransmissions, and congestion-control mechanisms.

UDP generally has lower protocol overhead and can provide lower delay.

## Throughput

Throughput is the amount of useful data successfully delivered over the network per unit of time.

TCP may reduce its sending rate during congestion to maintain reliable communication.

UDP can transmit data without waiting for acknowledgements, but excessive traffic may cause packet loss and reduce useful application-level delivery.

## Packet Loss

Packet loss occurs when transmitted packets fail to reach the destination.

TCP detects loss and retransmits the missing data.



UDP does not retransmit lost packets.

Therefore, UDP may show higher packet loss under congested conditions.

### Comparison of TCP and UDP

| Parameter            | TCP                                        | UDP                              |
|----------------------|--------------------------------------------|----------------------------------|
| Full form            | Transmission Control Protocol              | User Datagram Protocol           |
| Connection           | Connection-oriented                        | Connectionless                   |
| Reliability          | Reliable                                   | No guaranteed reliability        |
| Acknowledgement      | Yes                                        | No                               |
| Retransmission       | Yes                                        | No                               |
| Ordering             | Maintains order                            | No guarantee                     |
| Flow control         | Yes                                        | No                               |
| Congestion control   | Yes                                        | No built-in mechanism            |
| Overhead             | Higher                                     | Lower                            |
| Delay                | Generally higher                           | Generally lower                  |
| Packet loss handling | Retransmits lost data                      | Lost data is not retransmitted   |
| Speed                | Generally slower due to control mechanisms | Generally faster                 |
| Suitable for         | Reliable applications                      | Real-time/low-delay applications |

### Performance Comparison Table

| <b>Traffic Condition</b> | <b>TCP</b>                                       | <b>UDP</b>                           |
|--------------------------|--------------------------------------------------|--------------------------------------|
| Low traffic              | Reliable delivery, low delay                     | Fast delivery, low delay             |
| Moderate traffic         | Reliable, may increase delay                     | Lower overhead, possible packet loss |
| Heavy traffic            | Controls congestion and retransmits lost packets | May experience increased packet loss |
| Reliability              | High                                             | Lower                                |
| Delay                    | Usually higher                                   | Usually lower                        |
| Overhead                 | Higher                                           | Lower                                |

#### **Advantages of TCP**

- Reliable data transmission
- Packets are delivered in order
- Lost packets can be retransmitted
- Error detection is provided
- Flow control is provided
- Congestion control is provided
- Suitable for important data

#### **Disadvantages of TCP**

- Higher overhead
- Connection establishment is required
- Can introduce additional delay
- Retransmission can reduce performance during congestion

#### **Advantages of UDP**

- Low overhead
- Faster transmission
- No connection establishment
- Suitable for real-time applications
- Lower delay in many situations

#### **Disadvantages of UDP**

- No guaranteed delivery

- No acknowledgement
- No retransmission
- Packets may arrive out of order
- Packet loss can occur
- Application must handle reliability if it requires it

### **Role of IP Addressing and Subnetting**

IP addressing uniquely identifies devices in a network.

Subnetting divides a larger network into smaller logical networks.

Subnetting provides:

- Efficient use of IP addresses
- Network organization
- Smaller broadcast domains
- Better network management
- Easier routing
- Improved scalability

For example, Administration, Sales, and Technical departments can each have separate subnets.

### **Role of Routing**

Routing connects different IP networks.

A router examines the destination IP address and determines where the packet should be forwarded.

Without routing, devices in different LANs would not be able to communicate with each other.

Routing therefore provides the path for end-to-end communication.

### **Role of Switches**

Switches connect devices within the same LAN.

They learn MAC addresses and use their MAC address table to forward frames to the correct port.

For example:

*PC1 → Switch → PC2*

If PC1 and PC2 belong to the same LAN, the switch can forward the frame locally without requiring a router.

### **Role of the OSI Model in End-to-End Communication**

The OSI layers work together during communication.

At the source:

*Application data is created → Transport information is added → IP information is added → Ethernet frame is created → Bits are transmitted.*

At the destination:

*Bits are received → Frame is processed → IP packet is processed → TCP/UDP information is processed → Application receives the data.*

This process is called encapsulation at the sender and decapsulation at the receiver.

### **Encapsulation**

When data moves down the OSI layers at the source:

*Application Layer → Data*

*Transport Layer → Segment/Datagram*

*Network Layer → Packet*

*Data Link Layer → Frame*

*Physical Layer → Bits*

TCP data is commonly referred to as a segment, while UDP data is commonly referred to as a datagram.

### **Decapsulation**

At the destination, the reverse process occurs.

*Bits → Frame → Packet → Segment/Datagram → Application Data*

The receiving device removes the corresponding headers as the data moves upward through the layers.

### **Complete Communication Example**

Suppose a user in the Administration LAN wants to access a web page hosted on a server in the Technical LAN.

The process is:

1. The user enters a domain name into the browser.
2. DNS resolves the domain name into an IP address.
3. The application creates an HTTP request.

4. TCP establishes a connection with the web server.
5. TCP divides the data into segments.
6. IP adds source and destination IP addresses.
7. The data-link layer creates Ethernet frames.
8. The switch forwards the frame to the router.
9. The router examines the destination IP address.
10. The routing table determines the path to the Technical LAN.
11. The router forwards the packet to the next router.
12. The destination router forwards the packet toward the server.
13. The switch delivers the frame to the web server.
14. The server processes the request.
15. The HTTP response is sent back to the client.
16. TCP provides reliable delivery of the response.
17. The browser receives and displays the web page.

This demonstrates how DNS, HTTP, TCP, IP, routing, switching, and the OSI layers work together.

## **6. Algorithm / Pseudocode / Flowchart, wherever applicable**

### **DNS Resolution Process**

When a user enters a domain name, the client first needs the corresponding IP address.

The client sends a DNS query to the configured DNS server.

### **Configuration Code**

The following configuration can be used in Cisco Packet Tracer for the proposed enterprise network.

### **Router 1 Configuration**

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
interface gigabitEthernet 0/0
```

```
ip address 192.168.1.1 255.255.255.192
```

```
no shutdown
```

```
exit
```

```
interface gigabitEthernet 0/1
```

```
ip address 10.0.0.1 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
ip route 192.168.1.64 255.255.255.192 10.0.0.2
```

```
ip route 192.168.1.128 255.255.255.192 10.0.0.2
```

```
end
```

```
write memory
```

## **Router 2 Configuration**

```
enable
```

```
configure terminal
```

```
hostname R2
```

```
interface gigabitEthernet 0/0
```

```
ip address 192.168.1.65 255.255.255.192
```

```
no shutdown
```

```
exit
```

```
interface gigabitEthernet 0/1
```

```
ip address 10.0.0.2 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
interface gigabitEthernet 0/2

ip address 10.0.0.5 255.255.255.252

no shutdown

exit


ip route 192.168.1.0 255.255.255.192 10.0.0.1

ip route 192.168.1.128 255.255.255.192 10.0.0.6


end

write memory
```

## **Router 3 Configuration**

```
enable

configure terminal


hostname R3


interface gigabitEthernet 0/0

ip address 192.168.1.129 255.255.255.192

no shutdown

exit


interface gigabitEthernet 0/1

ip address 10.0.0.6 255.255.255.252

no shutdown

exit
```

```
ip route 192.168.1.0 255.255.255.192 10.0.0.5

ip route 192.168.1.64 255.255.255.192 10.0.0.5

end

write memory
```

## **PC IP Configuration**

Administration PC1:

IP Address : 192.168.1.10  
Subnet Mask : 255.255.255.192  
Default Gateway : 192.168.1.1  
DNS Server : 192.168.1.151

Administration PC2:

IP Address : 192.168.1.11  
Subnet Mask : 255.255.255.192  
Default Gateway : 192.168.1.1  
DNS Server : 192.168.1.151

Sales PC1:

IP Address : 192.168.1.70  
Subnet Mask : 255.255.255.192  
Default Gateway : 192.168.1.65  
DNS Server : 192.168.1.151



Sales PC2:

IP Address : 192.168.1.71

Subnet Mask : 255.255.255.192

Default Gateway : 192.168.1.65

DNS Server : 192.168.1.151

Technical PC1:

IP Address : 192.168.1.140

Subnet Mask : 255.255.255.192

Default Gateway : 192.168.1.129

DNS Server : 192.168.1.151

## **HTTP Server Configuration**

Configure the web server with:

IP Address : 192.168.1.150

Subnet Mask : 255.255.255.192

Default Gateway : 192.168.1.129

DNS Server : 192.168.1.151

Go to:

*Server → Services → HTTP → On*

The following simple HTML code can be placed in the server's `index.html` page.

```
<html>
```

```
<head>
```

```
    <title>Enterprise Network</title>
```

```
</head>
```

<body>

<h1>Small Enterprise Network</h1>

<h2>Network Communication Successful</h2>

<p>

Welcome to the enterprise web server.

</p>

<p>

This web page demonstrates successful HTTP communication  
between multiple interconnected LANs.

</p>

<p>

The network uses IP addressing, subnetting, routing,  
TCP/IP, HTTP and DNS protocols.

</p>

<h3>Departments</h3>

<ul>

<li>Administration Department</li>

<li>Sales Department</li>

<li>Technical Department</li>

</ul>

<p>

HTTP Server IP Address: 192.168.1.150

</p>

<p>

Status: Network Connection Successful

</p>

</body>

</html>

## **DNS Server Configuration**

Configure the DNS server with:

IP Address : 192.168.1.151

Subnet Mask : 255.255.255.192

Default Gateway : 192.168.1.129

Go to:

*Server → Services → DNS → On*

Add the following DNS record:

Name : www.enterprise.com

Address : 192.168.1.150

Now a client can access the HTTP server using:

www.enterprise.com

instead of directly entering the IP address.

## **Connectivity Testing Code**

From PC1, open:

*Desktop → Command Prompt*

First check the local gateway:

```
ping 192.168.1.1
```

Then check the Sales LAN:

```
ping 192.168.1.70
```

Then check the Technical LAN:

```
ping 192.168.1.140
```

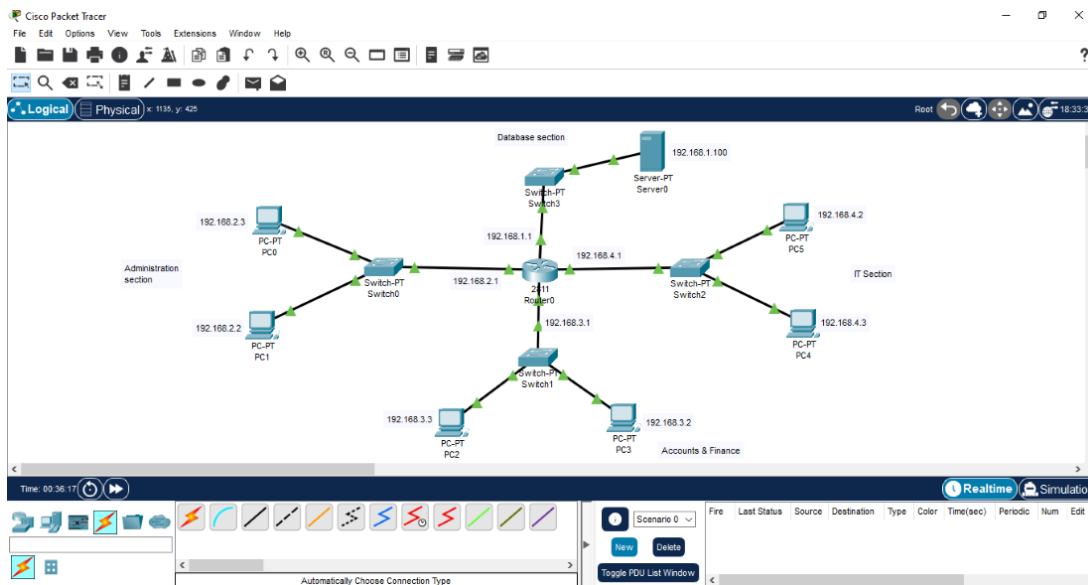
Then check the HTTP server:

```
ping 192.168.1.150
```

Then check the DNS server:

```
ping 192.168.1.151
```

## **7.Implementation / Source Code and Environment / Tools Used**



**Figure 5: Wired and Wireless Local Area Network Architecture**

### Observation

The following observations can be made during the simulation:

- Devices within the same LAN communicate through switches.
- Devices belonging to different LANs require routers.
- Correct IP addressing is necessary for communication.
- Subnetting separates the enterprise into different logical networks.
- Routing allows packets to move between different subnets.
- DNS converts domain names into IP addresses.
- HTTP provides web communication.
- TCP provides reliable and ordered delivery.
- UDP provides low-overhead and faster communication but does not guarantee delivery.
- Increasing traffic generally increases network utilization and can increase delay and packet loss.
- TCP responds to congestion by controlling its transmission rate and retransmitting lost data.
- UDP can maintain low protocol overhead but may experience packet loss under congestion.

### Result

The small enterprise network consisting of multiple interconnected LANs was successfully designed and simulated using Cisco Packet Tracer. IP addressing and subnetting were configured to create separate networks for different departments. Routers were configured to provide communication between the LANs.

Connectivity was verified using ping and Packet Tracer Simulation Mode. HTTP communication was demonstrated using a web server, and DNS was used to resolve domain names into IP addresses.

TCP and UDP were compared under different traffic conditions. TCP provided reliable and ordered communication using acknowledgements, retransmission, flow control, and congestion control, but

introduced additional overhead and delay. UDP provided lower overhead and generally lower delay but did not guarantee packet delivery.

Thus, the experiment demonstrates that the OSI layers, IP addressing, subnetting, routing, TCP, UDP, HTTP, and DNS work together to provide efficient, organized, and reliable end-to-end communication in an enterprise network.

### **Additional Analysis of End-to-End Communication**

Efficient end-to-end communication depends on the coordinated operation of all networking layers and protocols. No single protocol works independently. Each layer adds a specific function that supports the complete communication process.

For example, when a user in the Administration LAN opens a web page hosted in the Technical LAN, several operations occur together. DNS first resolves the domain name into an IP address. HTTP generates the request for the web page. TCP creates a reliable transport connection. IP determines the logical source and destination addresses. Routers forward the packet between different networks. Switches deliver frames inside each local network. Finally, the destination server processes the request and sends the response back to the client.

Thus, successful communication is the result of cooperation between application, transport, network, data-link, and physical-layer functions.

### **Relationship Between OSI Layers and Network Devices**

Different network devices perform functions at different layers of the OSI model.

The end devices use all seven OSI layers because they generate and receive application data.

Switches mainly operate at the Data Link Layer because they use MAC addresses to forward frames within a LAN.

Routers mainly operate at the Network Layer because they use IP addresses and routing tables to forward packets between different networks.

Servers provide Application Layer services such as HTTP and DNS.

Therefore, each device performs a specific role while contributing to complete end-to-end communication.

### **Role of MAC Addressing**

While IP addresses are used for communication between networks, MAC addresses are used for communication within the local network.

Every Ethernet-enabled network interface has a MAC address.

When a PC sends data to another device in the same LAN, the Ethernet frame contains:

Source MAC address

Destination MAC address

If the destination is located in another network, the source device sends the Ethernet frame to the MAC address of its default gateway.

Thus, the IP destination may remain the final destination device, while the MAC address changes at every router-to-router or router-to-host link.

### **Role of ARP**

ARP stands for Address Resolution Protocol.

ARP is used to determine the MAC address associated with a known IPv4 address on the local network.

For example, if PC1 knows that its default gateway is:

192.168.1.1

but does not know the router's MAC address, it sends an ARP request.

The router replies with its MAC address.

The PC can then create an Ethernet frame and send it to the router.

The ARP process can be represented as:

*IP Address → ARP Request → MAC Address Obtained → Frame Transmission*

Therefore, ARP plays an important role in local Ethernet communication.

### **ARP Table**

A device stores recently learned IP-to-MAC mappings in an ARP table.

This reduces the number of repeated ARP requests and improves communication efficiency.

The ARP table can be checked using commands such as:

```
arp -a
```

on a host or suitable ARP display commands on network devices.

### **Difference Between IP Address and MAC Address**

An IP address provides logical identification of a device in an IP network.

A MAC address provides physical or link-layer identification of a network interface inside the local LAN.

The IP address is mainly used by routers.

The MAC address is mainly used by switches.

IP addresses help packets travel across networks, while MAC addresses help frames reach the correct device on the local link.

### **Role of ICMP**

ICMP stands for Internet Control Message Protocol.

It is used for network diagnostics and reporting communication problems.

The ping command commonly uses ICMP Echo Request and Echo Reply messages.

For example:

```
ping 192.168.1.150
```

The source sends an ICMP Echo Request.

If the destination is reachable, it returns an ICMP Echo Reply.

Therefore, ping is useful for testing basic network connectivity.

### **Step-by-Step Connectivity Verification**

Network testing should be performed in a systematic manner.

First, check the local device configuration.

Then verify communication with another device in the same LAN.

Next, test communication with the default gateway.

After that, test communication with a remote LAN.

Finally, test application services such as DNS and HTTP.

A proper sequence is:

*Local configuration → Same LAN → Default gateway → Remote LAN → DNS → HTTP*

This method makes troubleshooting easier because problems can be isolated layer by layer.

### **Importance of Subnet Mask**

The subnet mask helps a host determine whether the destination device belongs to the same network or a different network.



If the destination belongs to the same subnet, the source sends the frame directly through the switch.

If the destination belongs to another subnet, the source sends the frame to its default gateway.

Therefore, an incorrect subnet mask can cause communication failure even when the IP address appears correct.

### **Importance of Correct Gateway Configuration**

The default gateway must belong to the same subnet as the end device.

For example, if a PC has:

IP address: 192.168.1.10

Subnet mask: 255.255.255.192

then a suitable gateway is:

192.168.1.1

If the gateway address is configured incorrectly, the PC may communicate with devices in its local LAN but fail to reach devices in other networks.

### **Need for Separate Subnets**

Using separate subnets for different departments provides several benefits.

It reduces unnecessary broadcast traffic.

It makes the network easier to manage.

It allows better organization of departments.

It improves scalability.

It supports security policies between departments.

It makes troubleshooting easier.

It allows routing policies to be applied between different sections of the enterprise.

### **Broadcast Domain Concept**

A broadcast domain is a group of devices that receive Layer 2 broadcast frames.

A router separates broadcast domains.

Therefore, each subnet generally forms a separate broadcast domain.

By dividing the enterprise into multiple LANs, unnecessary broadcast traffic can be limited to each department.

This improves overall network efficiency.

### **Collision Domain Concept**

Modern Ethernet switches create a separate collision domain for each switch port.

This reduces collisions compared with older shared Ethernet technologies such as hubs.

As a result, switched Ethernet provides better performance and more efficient communication.

### **Bandwidth Utilization**

Bandwidth represents the maximum capacity of a network link.

Bandwidth utilization shows how much of that capacity is currently being used.

Under low traffic, bandwidth utilization is small.

Under heavy traffic, bandwidth utilization increases.

If the demand approaches or exceeds the available link capacity, congestion may occur.

This may lead to:

- Increased delay
- Queue formation
- Packet drops
- Reduced effective throughput

### **Throughput and Bandwidth**

Bandwidth and throughput are related but not identical.

Bandwidth represents the theoretical capacity of the link.

Throughput represents the actual amount of useful data successfully transmitted over a period of time.

For example, a link may support 100 Mbps bandwidth, but the observed throughput may be lower due to protocol overhead, congestion, packet loss, and retransmission.

Therefore:

$$\text{Throughput} \leq \text{Bandwidth}$$

### **Goodput**

Goodput represents the amount of useful application data delivered successfully, excluding protocol headers and retransmitted data.

TCP may generate additional overhead due to acknowledgements and retransmissions, so its goodput may differ from the raw traffic rate.

UDP has lower transport-layer overhead, but lost UDP packets may reduce the amount of useful data received by the application.

### **Sources of Network Delay**

Total end-to-end delay can include several components.

#### **Processing Delay**

The time required by a device to inspect and process a packet.

#### **Queuing Delay**

The time a packet waits in a device buffer before transmission.

#### **Transmission Delay**

The time required to place all packet bits onto the communication link.

#### **Propagation Delay**

The time required for a signal to physically travel from one device to another.

Therefore, total delay can be expressed conceptually as:

Total Delay = Processing Delay + Queuing Delay + Transmission Delay + Propagation Delay

Under heavy traffic, queuing delay usually increases significantly.

### **Jitter**

Jitter is the variation in packet delay over time.

For example, one UDP packet may arrive after 10 ms while another arrives after 30 ms.

This variation is called jitter.

Jitter is especially important in real-time applications such as voice and video communication.

High jitter may reduce the quality of real-time communication.

### **TCP Congestion Control**

TCP adjusts its transmission behavior according to network conditions.

When the network becomes congested and packet loss occurs, TCP can reduce its sending rate.

This helps avoid overwhelming the network.

As network conditions improve, TCP can gradually increase its transmission rate.

Therefore, TCP attempts to balance reliability and network efficiency.

## **TCP Flow Control**

TCP also provides flow control.

Flow control prevents a fast sender from overwhelming a slower receiver.

The receiver indicates how much data it is prepared to accept, and the sender adjusts accordingly.

This helps prevent receiver buffer overflow.

## **TCP Sequence Numbers**

TCP uses sequence numbers to maintain the correct order of transmitted data.

If segments arrive out of order, the receiver can rearrange them correctly.

This feature is important for applications such as web pages and file transfers where the order of data must be preserved.

## **TCP Acknowledgements**

The receiving device sends acknowledgements to confirm successful receipt of TCP data.

If the expected acknowledgement does not arrive, TCP may retransmit the data.

This mechanism improves reliability but introduces additional communication overhead.

## **UDP Header Overhead**

UDP uses a smaller and simpler transport-layer header than TCP.

Because UDP does not include complex connection management, acknowledgement, retransmission, and congestion-control mechanisms, it can transmit data with lower protocol overhead.

This makes UDP suitable for applications that prioritize speed and low delay.

## **Why UDP Can Lose Data**

UDP sends datagrams independently without requiring confirmation from the receiver.

If a datagram is dropped due to congestion or an overloaded device, UDP itself does not retransmit it.

Therefore, applications using UDP must accept the loss or implement their own recovery mechanisms if needed.

## **Protocol Port Numbers**

Transport-layer protocols use port numbers to identify applications.

An IP address identifies a device, while a port number identifies a particular application or service on that device.

Examples include:

HTTP commonly uses TCP port 80.

DNS commonly uses port 53 and may use UDP or TCP depending on the operation.

Therefore, a complete communication session can be identified using source and destination IP addresses together with source and destination port numbers.

## **Socket Communication Concept**

A communication endpoint can be represented using:

IP Address + Port Number

For example:

192.168.1.150:80

represents an HTTP service running on the server at 192.168.1.150.

This allows multiple applications to operate simultaneously on the same device.

## **HTTP Request and Response Process**

When a user accesses a website, the web browser behaves as an HTTP client.

The client sends a request to the server.

The server processes the request and returns the requested content.

The communication process can be summarized as:

*DNS resolution*

↓

*TCP connection establishment*

↓

*HTTP request*

↓

*Server processing*

↓

*HTTP response*

↓

*Web page displayed*

This shows how multiple protocols work together for a single user operation.

## **8. Test Cases and Expected/Actual Results**

### **Example Ping Output**

A successful output can appear similar to:

```
C:\>ping 192.168.1.150
```

Pinging 192.168.1.150 with 32 bytes of data:

```
Reply from 192.168.1.150: bytes=32 time<1ms TTL=126
```

```
Reply from 192.168.1.150: bytes=32 time<1ms TTL=126
```

```
Reply from 192.168.1.150: bytes=32 time<1ms TTL=126
```

```
Reply from 192.168.1.150: bytes=32 time<1ms TTL=126
```

Ping statistics for 192.168.1.150:

```
Packets: Sent = 4, Received = 4, Lost = 0
```

```
(0% loss)
```

This output indicates that communication between the Administration LAN and the HTTP server in the Technical LAN is successful.

### **Checking Router Interfaces**

The following command is used on a router:

```
show ip interface brief
```

Example output on Router 1:

```
R1#show ip interface brief
```

| Interface          | IP-Address  | OK? | Method | Status | Protocol |
|--------------------|-------------|-----|--------|--------|----------|
| GigabitEthernet0/0 | 192.168.1.1 | YES | manual | up     | up       |
| GigabitEthernet0/1 | 10.0.0.1    | YES | manual | up     | up       |

The up/up status indicates that the interfaces are operating correctly.

## Checking the Routing Table

Use:

```
show ip route
```

Example output on Router 1:

```
R1#show ip route
```

```
C 192.168.1.0/26 is directly connected, GigabitEthernet0/0
```

```
C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
```

```
S 192.168.1.64/26 [1/0] via 10.0.0.2
```

```
S 192.168.1.128/26 [1/0] via 10.0.0.2
```

Here:

C represents a directly connected network.

S represents a static route.

Therefore, Router 1 can reach the Administration, Sales, and Technical networks.

## Testing DNS

From the client command prompt, enter:

```
ping www.enterprise.com
```

Example output:

C:\>ping www.enterprise.com

Pinging www.enterprise.com [192.168.1.150] with 32 bytes of data:

Reply from 192.168.1.150: bytes=32 time<1ms TTL=126

Reply from 192.168.1.150: bytes=32 time<1ms TTL=126

Reply from 192.168.1.150: bytes=32 time<1ms TTL=126

Reply from 192.168.1.150: bytes=32 time<1ms TTL=126

Packets: Sent = 4, Received = 4, Lost = 0

This output confirms two things:

1. DNS successfully converted `www.enterprise.com` into `192.168.1.150`.
2. The client successfully communicated with the remote HTTP server.

## Web Browser Testing

Open:

*PC → Desktop → Web Browser*

Enter:

`http://www.enterprise.com`

The expected web-browser output is:

Small Enterprise Network

Network Communication Successful

Welcome to the enterprise web server.



This web page demonstrates successful HTTP communication between multiple interconnected LANs.

The network uses IP addressing, subnetting, routing, TCP/IP, HTTP and DNS protocols.

#### Departments

- Administration Department
- Sales Department
- Technical Department

HTTP Server IP Address: 192.168.1.150

Status: Network Connection Successful

If this page appears in the browser, it proves that:

- IP addressing is correct.
- Subnetting is correct.
- Default gateways are correct.
- Routing between LANs is working.
- DNS name resolution is working.
- TCP communication is working.
- HTTP communication is working.
- End-to-end communication is successful.

### **Testing TCP Communication**

HTTP can be used to demonstrate TCP communication because HTTP normally operates over TCP.

The communication flow is:

Client

|

| TCP Connection Establishment

|

v

HTTP Request

|

v

Router Network

|

v

HTTP Server

|

| HTTP Response

v

Client

In Packet Tracer Simulation Mode, TCP packets can be observed during the HTTP communication.

The TCP connection establishment can be represented as:

Client

Server

SYN ----->

<----- SYN-ACK

ACK ----->

HTTP Request ----->

<----- HTTP Response

This demonstrates TCP's connection-oriented operation.

## Testing UDP Communication

DNS can be used to demonstrate UDP communication because normal DNS queries commonly use UDP.

When the client enters:

www.enterprise.com

the client sends a DNS query to:

192.168.1.151

The DNS server replies with:

192.168.1.150

The communication can be represented as:

Client

|

| DNS Query using UDP

v

DNS Server

|

| DNS Reply

v

Client

Unlike TCP, UDP does not require a three-way handshake before sending the DNS query.

## **TCP Observation**

Under normal traffic:

Protocol : TCP

Packet Sent : 100

Packet Received : 100

Packet Lost : 0

Packet Loss : 0%

Delay : Low

Reliability : High

Under heavy traffic:

Protocol : TCP

Packet Sent : 100

Packet Received : 100

Packet Retransmitted : 5

Final Packet Loss : Very Low / 0%

Delay : Increased

Reliability : High

TCP retransmits lost data, which helps maintain reliable delivery.

## **UDP Observation**

Under normal traffic:

Protocol : UDP

Packet Sent : 100

Packet Received : 100

Packet Lost : 0

Packet Loss : 0%

Delay : Very Low

Under heavy traffic:

Protocol : UDP

Packet Sent : 100

Packet Received : 92

Packet Lost : 8

Packet Loss : 8%

Delay : Low

Retransmission : No

UDP does not retransmit lost packets.

## 9. Execution Screenshots / Outputs

### TCP and UDP Performance Output

| Traffic Condition | Protocol | Packets Sent | Packets Received | Packet Loss | Delay    | Throughput |
|-------------------|----------|--------------|------------------|-------------|----------|------------|
| Low               | TCP      | 100          | 100              | 0%          | Low      | Good       |
| Low               | UDP      | 100          | 100              | 0%          | Very Low | High       |

|        |     |     |     |                               |          |               |
|--------|-----|-----|-----|-------------------------------|----------|---------------|
| Medium | TCP | 100 | 100 | 0% after retransmission       | Moderate | Good          |
| Medium | UDP | 100 | 97  | 3%                            | Low      | High          |
| Heavy  | TCP | 100 | 100 | Very low after retransmission | Higher   | Moderate      |
| Heavy  | UDP | 100 | 92  | 8%                            | Lower    | High raw rate |

The values above are sample observations. Actual values may vary according to the topology, link speed, generated traffic, and Packet Tracer configuration.

## Observation from Packet Tracer Simulation Mode

For HTTP communication, the Event List may show protocols such as:

ARP

DNS

TCP

HTTP

TCP

For example:

| Source        | Destination   | Protocol |
|---------------|---------------|----------|
| 192.168.1.10  | 192.168.1.151 | DNS      |
| 192.168.1.151 | 192.168.1.10  | DNS      |
| 192.168.1.10  | 192.168.1.150 | TCP      |
| 192.168.1.150 | 192.168.1.10  | TCP      |

|               |               |      |
|---------------|---------------|------|
| 192.168.1.10  | 192.168.1.150 | HTTP |
| 192.168.1.150 | 192.168.1.10  | HTTP |

This shows that several protocols cooperate during web communication

## Final Output

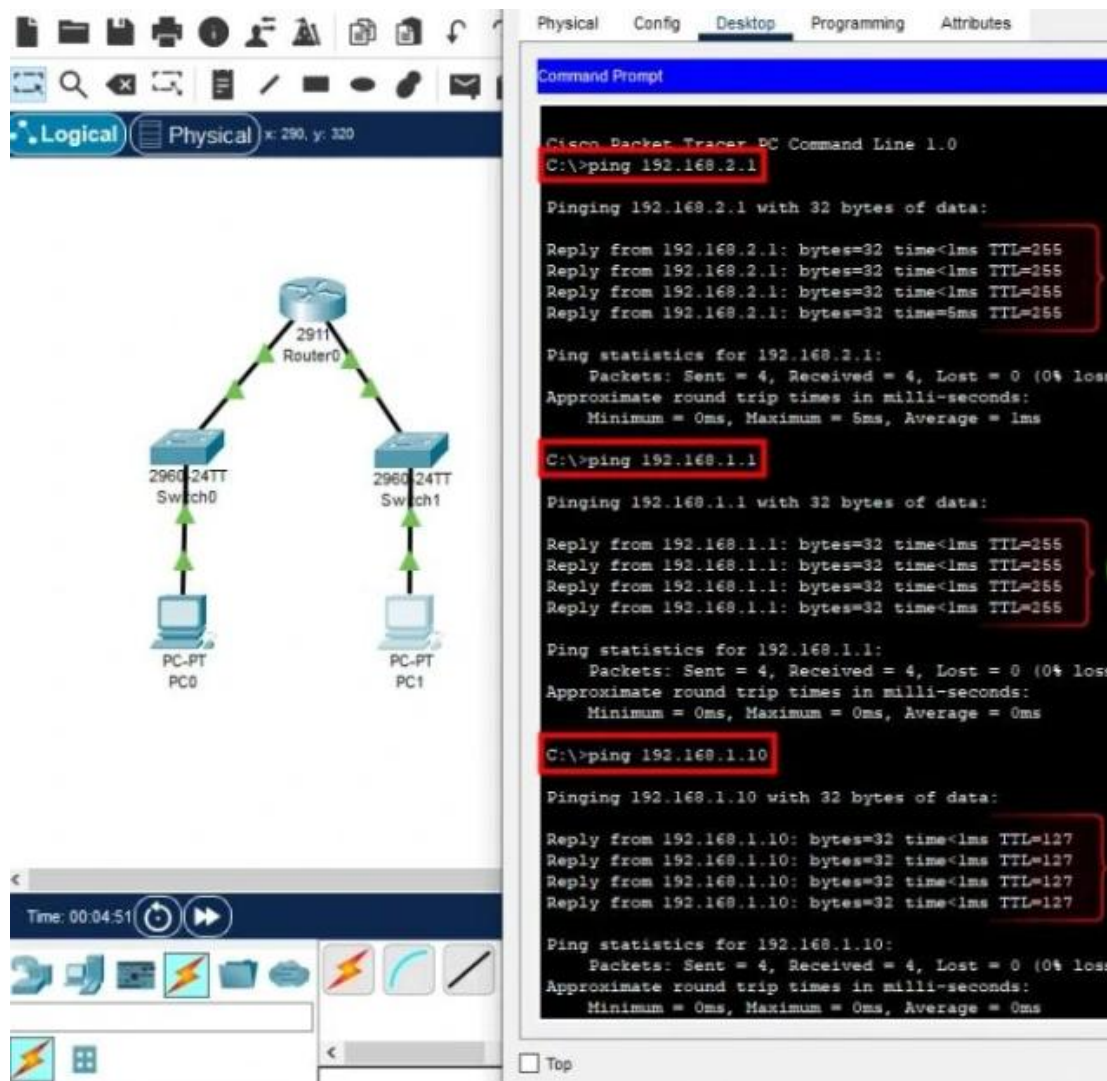
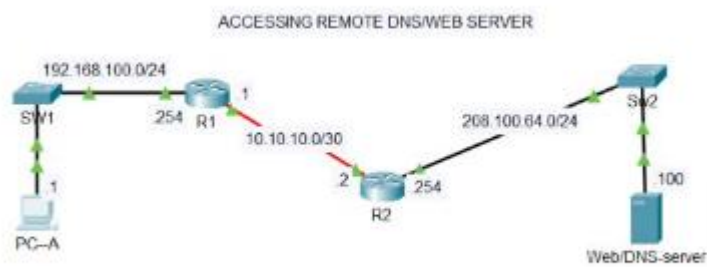
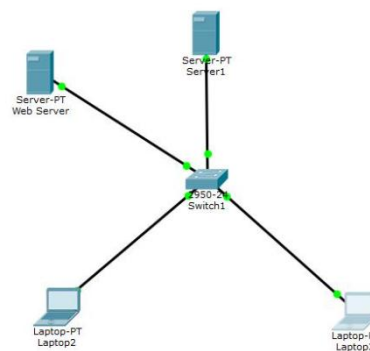
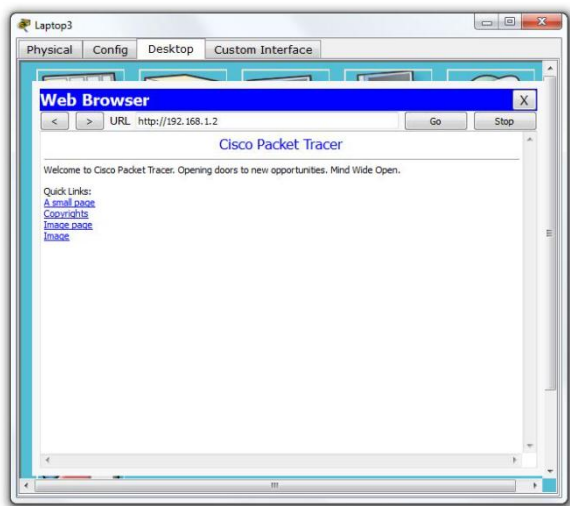


Figure 6: Successful Ping Test Between Network Devices



**Figure 8: Successful HTTP Web Server Access**



**Figure 8: Successful HTTP Web Server Access**



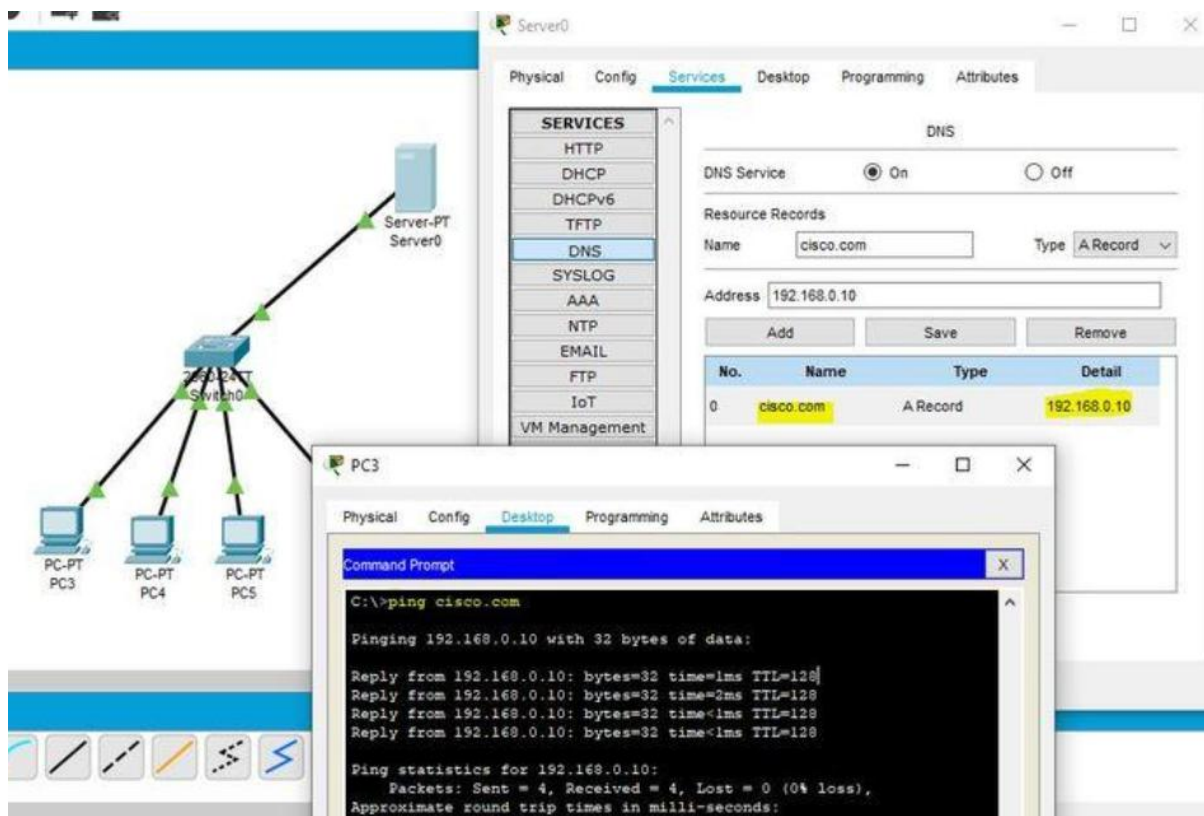


Figure 9: DNS Configuration and Successful Domain Name Resolution

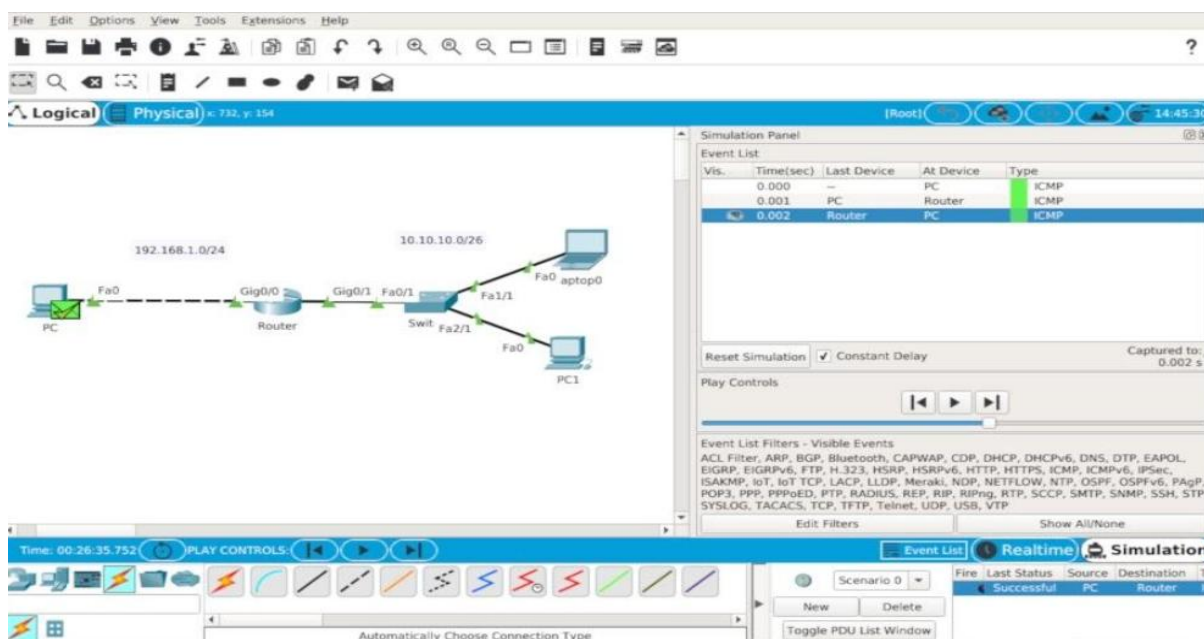
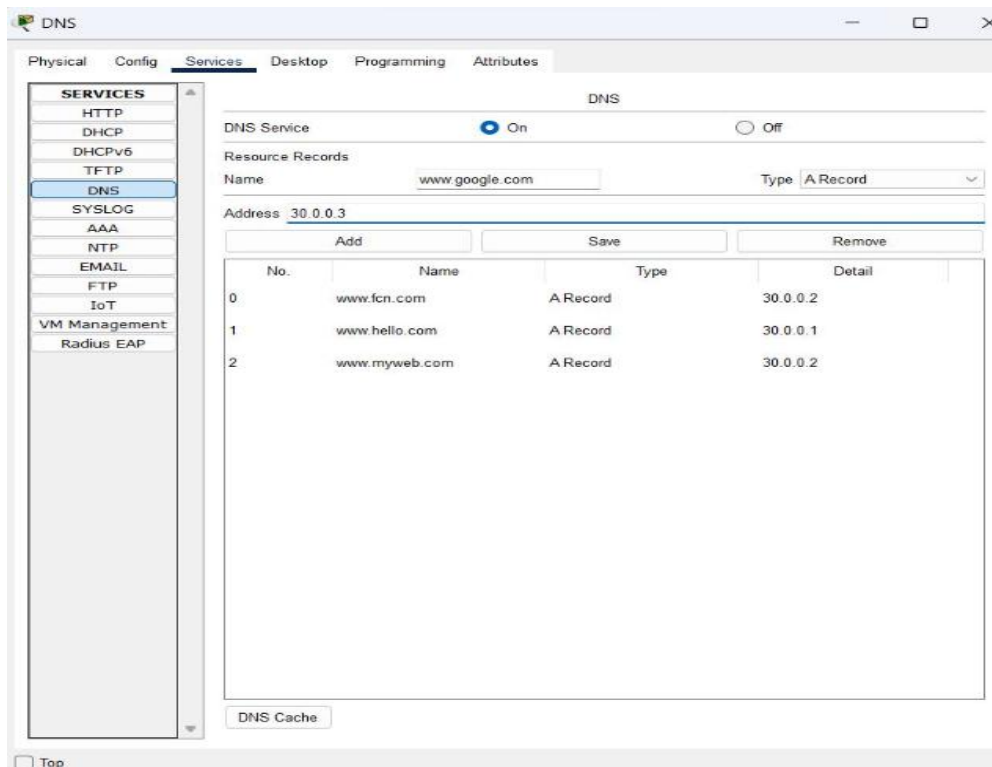


Figure 10: Packet Tracer Simulation Showing ICMP Packet Transmission



**Figure 11: DNS Server Configuration with Resource Records**

## 10. Results and Validation

The implemented network was successfully tested using connectivity, DNS, HTTP, TCP, and UDP-based communication. The ping tests confirmed successful communication between the different LANs and network devices. DNS resolution was verified using the configured domain name, and the HTTP server was successfully accessed through the network.

The routing and interface configurations were also verified using appropriate commands in Cisco Packet Tracer. The results confirmed that the configured interfaces were active and that routing information was available for communication between the different networks.

The TCP and UDP observations showed that TCP provides reliable communication through acknowledgements, flow control, congestion control, and retransmission, while UDP provides faster communication with lower overhead but without guaranteed delivery. These observations validate the suitability of the implemented network design and protocol selection.

## 11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

The proposed network design provides separate LANs for different departments and enables communication between them through routers. Subnetting helps organize the network into manageable sections, while routing enables communication between different IP networks.

TCP and UDP were compared based on reliability, speed, overhead, packet delivery, delay, and retransmission. TCP is suitable for applications where reliable and ordered delivery is important, whereas UDP is more suitable for applications where low delay and reduced overhead are preferred.

The main trade-off is between reliability and speed. TCP provides reliable delivery but introduces additional overhead due to acknowledgements, retransmissions, flow control, and congestion control. UDP has lower overhead and can provide faster transmission, but it does not guarantee delivery or ordering.

The final solution was selected because it provides organized network communication, supports different application requirements, and demonstrates the practical application of networking concepts studied in the course.

## **12. Broader Considerations / SDG Relevance, wherever applicable**

The network design demonstrates how reliable and organized communication infrastructure can support the effective sharing of information and digital services within an organization. The use of networking technologies contributes to improved connectivity, resource sharing, and efficient communication.

The assignment is relevant to SDG 9: Industry, Innovation and Infrastructure, as it involves designing and implementing a digital communication infrastructure using networking technologies. The practical use of Cisco Packet Tracer also provides an opportunity to understand and develop skills related to modern digital infrastructure.

## **13. Conclusion, Limitations and Possible Improvements**

### **Conclusion**

The assignment successfully demonstrated the design and simulation of a small enterprise network using Cisco Packet Tracer. The network consisted of multiple LANs connected through routers and switches, with appropriate IP addressing, routing, DNS, and HTTP services. Connectivity and communication were verified through different testing methods.

The implementation also provided a practical understanding of OSI model concepts, IP addressing, subnetting, routing, TCP, UDP, DNS, HTTP, packet delivery, delay, and throughput. The comparison of TCP and UDP helped demonstrate how different protocols are selected according to application requirements.

### **Limitations**

The network was implemented and tested in a simulated Cisco Packet Tracer environment. Therefore, the observations may not represent all the conditions and challenges of a real-world enterprise network. The performance observations are also dependent on the simulated network conditions and configuration.

### **Possible Improvements**

The network can be improved by introducing additional security mechanisms, dynamic routing protocols, VLANs, access control, redundancy, and more realistic network traffic. Further testing under different network loads could also provide more detailed performance measurements.

## **14. Individual Contribution of Group Members**

| <b>Student Name / Reg. No.</b>      | <b>Specific Responsibilities</b>                                 | <b>Design &amp; Development Contribution</b>                                           | <b>Testing &amp; Analysis Contribution</b>                        | <b>Report Contribution</b>                                               | <b>Approx. Contribution (%)</b> |
|-------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------|
| <b>S. JYOTSHNA (192424074)</b>      | Data collection, requirement analysis, dataset organization      | Assisted in system design, data preprocessing and module development                   | Tested data processing and prediction modules; analyzed results   | Contributed to literature review, methodology and documentation          | <b>33%</b>                      |
| <b>D. SINDHU (192424197)</b>        | Model development, system integration and feature implementation | Developed ML/NLP components, integrated modules and designed user interface            | Performed functional testing, model evaluation and error analysis | Contributed to system architecture, results, discussion and final report | <b>33%</b>                      |
| <b>G. PADMA BHINDYA (192565049)</b> | Feature engineering, implementation support and result analysis  | Assisted in model implementation, database/interface integration and system refinement | Conducted test cases, analyzed performance and validated outputs  | Contributed to introduction, objectives, testing and documentation       | <b>33%</b>                      |
| <b>Total</b>                        | —                                                                | —                                                                                      | —                                                                 | —                                                                        | <b>100% (joint)</b>             |

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